① Aspect 14 (Most Central)

Make generalisations based on the results of applying mathematical procedures

The core requirement of this problem is not to substitute values and compute a numerical answer, but to:

- Analyse the given formula

$$D = \frac{dv}{60n}$$

- Consider what happens when n **is doubled while** d **and** v **remain unchanged**
- Derive a **general conclusion** about how the variables are related

The correct conclusion is:

> When n is doubled, D becomes half of its original value.

This is a **proportional generalisation derived from the algebraic structure of the formula**, making Aspect 14 the most prominent.

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② Aspect 7

Manipulate numbers, algebraic expressions, and equations

Students must:

- Correctly interpret the fractional structure of the formula
- Recognise that n appears in the denominator
- Use algebraic reasoning to determine the effect of doubling a variable

Although no complex computation is required, this is a **clear example of manipulating algebraic relationships**, rather than performing routine arithmetic.

③ Aspect 5

Understand and utilise constructs based on definitions, rules, and formal systems

Solving the problem depends on understanding:

- The definitions of the variables (d, v, n, D)
- The basic rule that increasing the denominator of a fraction decreases its value
- The concept of inverse proportionality

The reasoning relies on **formal algebraic rules and variable relationships**, not on intuition alone.

④ Aspect 13

Use an understanding of the context to guide the mathematical solving process

The real-world nursing context supports interpretation by helping students understand that:

- The drip rate D represents drops per minute
- Increasing the infusion time means the fluid must flow more slowly

Although the conclusion follows directly from algebra, the context **reinforces the reasonableness and meaning of the result**.

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⑤ Aspect 8

Articulate a solution, showing or summarising mathematical reasoning

The instruction "*Describe precisely how D changes*" requires:

- Clear and accurate verbal explanation
- Precise language (e.g. "is halved," rather than "decreases")

Communicating the relationship between variables is therefore an important part of the task.

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⑥ Aspect 1

Perform a simple calculation

Some students may:

- Assume a specific value for n
- Double it
- Compare the corresponding values of D

However, such calculations are **supporting tools rather than the main goal** of the problem, so this aspect is less central.

① Aspect 14（最核心）

Make generalisations based on the results of applying mathematical procedures

📌 本题的核心并不是代入数值计算，而是要求学生：

- 分析公式

$$D = \frac{dv}{60n}$$

- 思考当 n **翻倍，而** d **和** v **保持不变时**，变量之间的关系如何变化
- 得出一个**一般性结论**（而非具体数值）

👉 结论是：

> 当 n 翻倍时，D 变为原来的一半。

这正是**基于代数结构得出的比例性概括**，体现了从公式推断变化规律的能力，因此 Aspect 14 是最突出的方面。

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② Aspect 7

Manipulate numbers, algebraic expressions and equations

📌 学生需要：

- 正确理解分式结构
- 理解 n 位于分母中
- 运用代数思维判断"翻倍"的影响

👉 虽然不需要复杂运算，但这是一个**典型的代数关系操纵问题**，而非单纯算术。

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③ Aspect 5

Understand and utilise constructs based on definitions, rules and formal systems

📌 本题依赖学生对以下内容的理解：

- 变量的定义（d, v, n, D）
- 分式中"分母变大 → 整体变小"的基本规则
- 比例与反比例关系

👉 解题依托的是**正式的代数规则和变量关系**，而非直觉猜测。

④ Aspect 13

Use an understanding of the context to guide the mathematical solving process

📌 现实情境（输液护理）帮助学生理解：

- 滴速 D 表示“每分钟滴数”
- 输液时间变长意味着“滴得更慢”

👉 虽然结论可由代数推出，但**现实背景强化了结果的合理性与可解释性**，体现了情境对数学理解的支持作用。

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⑤ Aspect 8

Articulate a solution, showing or summarising mathematical reasoning

📌 题目要求：

- “Describe precisely how D changes”
- 强调**语言表达的准确性**（例如“变为原来的一半”，而不是“变小”）

👉 清楚表达变量变化关系，是评价该题的重要部分。

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⑥ Aspect 1

Perform a simple calculation

📌 学生可能通过：

- 假设一个具体 n 值
- 再令 n 翻倍
- 比较前后 D 的值

👉 但这种计算只是**辅助理解的手段**，并非题目核心要求，因此相关性较低。